THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) $\frac{9}{2}$ **(b)** -2 **(c)** $-\frac{9}{2}$ **(d)** 2

2. The product of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) $-\frac{9}{2}$ **(b)** -2 **(c)** 2 **(d)** $\frac{9}{2}$

3. The roots of $x^2 - 11x + 18 = 0$ differ by: 1

(a) 7 **(b)** 11 **(c)** 8 **(d)** 18

4. If $x = 4$ is a root of $x^2 - 9x + k = 0$, then $k =$ 1

(a) 5 **(b)** 20 **(c)** -20 **(d)** 52

5. The equation $x^2 + 6x + 8 = 0$ has: 1

(a) two equal real roots **(b)** two distinct real roots

(c) three real roots **(d)** no real roots

6. The larger root of $x^2 - 11x + 24 = 0$ is: 1

(a) 11 **(b)** 8 **(c)** 3 **(d)** 24

7. The sum of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) $-\frac{7}{3}$ **(b)** $\frac{7}{3}$ **(c)** -2 **(d)** 2

8. The product of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) $\frac{7}{3}$ **(b)** $-\frac{7}{3}$ **(c)** 2 **(d)** -2

9. The roots of $x^2 - 2x - 15 = 0$ differ by: 1

(a) 9 **(b)** 8 **(c)** 15 **(d)** 2

10. If $x = 5$ is a root of $x^2 - 11x + k = 0$, then $k =$ 1

(a) 80 **(b)** 6 **(c)** 30 **(d)** -30

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve for x: $2x^2 - 9x + 4 = 0$ by the quadratic formula. 2

12. If $x = 4$ is a root of $x^2 - 9x + k = 0$, find k and the other root. 2

13. Find k so that $kx^2 - 6x + 1 = 0$ has equal roots. 2

14. Solve for x: $3x^2 + 7x - 6 = 0$ by the quadratic formula. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{5}{6}$. **3**

16. Find the nature of the roots of $3x^2 - 7x + 2 = 0$ and, if real, find them. **3**

17. Find the roots of $x^2 - 10x + 21 = 0$ using the quadratic formula (if they exist). **3**

OR

Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{7}{12}$. **3**

18. Find the nature of the roots of $2x^2 + 5x + 4 = 0$ and, if real, find them. **3**

OR

Find the roots of $x^2 + 2x - 8 = 0$ using the quadratic formula (if they exist). **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A train travels 200 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 2 hours less for the journey. Find the original speed. **5**

OR

The altitude of a right triangle is 1 cm less than its base and the hypotenuse is 29 cm. Find the base and altitude. **5**

20. A train travels 280 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the journey. Find the original speed. **5**

OR

A motorboat goes 15 km upstream and 15 km downstream in 4 hours. If the speed of the stream is 2 km/h, find the speed of the boat in still water. **5**

END OF QUESTION PAPER — Total: 40 Marks

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